Lecture 1

Introduction

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- Notes about the course:
  - 4 problems (5% each).
  - Weekly quizzes (10% total).
  - 2 midterms (15% each).
  - Exam (40% total).
- Algebra is concerned with studying the implications that certain collections of axioms have on solving problems.

Definition 1: Let G be a set. A binary operator $*$ is a function $*: G \times G \to G$.

Definition 2: Let G be a set. G is a group under a binary operator $*$ if $(G, *)$ satisfies all 3 of the following criteria:

1. Associativity:
   - $\forall a, b, c \in G;\ (a * b) * c = a * (b * c)$.
2. Existence of an identity element:
   - $\exists e \in G;\ \forall a \in G;\ a * e = e * a = a$.
3. Existence of inverses:
   - $\forall a \in G;\ \exists b \in G;\ a * b = b * a = e$.

Definition 3: A group $(G, *)$ is called abelian if, in addition to satisfying the group axioms, $*$ satisfies the commutativity of $*$ on G:

$$\forall a, b \in G;\ a * b = b * a$$

Example 1: Which of the following are groups?

(a) $\mathbb{Z}$ under addition?

Solution:

Yes!

(b) $\mathbb{N}$ under addition?

Solution:

No! There is no inverse for any element except 0 if you let $0 \in \mathbb{N}$.

(c) $\mathbb{Z}$ under multiplication?

Solution:

No! Many elements, such as 2, do not have a multiplicative inverse in $\mathbb{Z}$.

(d) $\mathbb{Z}$ under subtraction?

Solution:

No! Subtraction isn't associative. For example:

$$(1-2)-3=-4 \neq 2=1-(2-3)$$

Example 2: Do any of the following sets under addition constitute a group?

$$\{1,2,3\}, \quad \{0,1,2,3\}, \quad \{-3,-2,-1,0,1,2,3\}$$

Solution:

No, since none of them are closed under addition. For example, $1+3=4$ does not belong to any of the sets.